

WaRee11 v1.5

User Manual and Workflow Guide

River Simulation and Hydraulic Modeling for iPad



Draft Manual
Prepared for WaRee11 Application Development

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Preface

This manual is a draft user guide for WaRee11 v1.5, an iPad-based river simulation and hydraulic modeling application. The goal of WaRee11 is to provide a clear, accessible, and field-friendly workflow for setting up a one-dimensional river model, defining river network geometry, entering cross-section data, configuring boundary conditions, running hydraulic simulations, checking diagnostics, and reviewing results.

Version 1.5 focuses on simplifying the user experience. Instead of presenting all technical information on a single crowded dashboard, the application uses a summary-first workflow. The Project Preview screen behaves as a Project Launcher, while detailed metadata, readiness checks, diagnostics, and project management functions are moved into dedicated detail screens.

Note

This document is a working draft. Screenshots, final button names, and final UI labels should be updated after the WaRee11 v1.5 implementation is finalized.

Revision History

Version	Date	Description
0.1	Draft	Initial WaRee11 v1.5 manual structure.

1 Introduction

1.1 What is WaRee11?

WaRee11 is a river simulation and hydraulic modeling application designed for iPad. It supports a simplified one-dimensional river modeling workflow inspired by professional hydraulic modeling tools such as MIKE 11, HEC-RAS, SOBEK, and D-Flow FM. The application is intended for students, researchers, engineers, and field users who need an intuitive environment for learning, preparing, and reviewing river model simulations.

1.2 Purpose of Version 1.5

The main objective of WaRee11 v1.5 is to improve usability by reducing screen clutter and organizing the modeling process into clear workflow stages. The application emphasizes:

- A clean Project Launcher screen.
- A separate Project Detail screen for full project information.
- Summary-first design for each modeling module.
- Clear readiness indicators before running simulations.
- Improved navigation between model setup, diagnostics, and results.
- A more professional iPad workflow suitable for demonstrations, teaching, and early field testing.

1.3 Target Users

WaRee11 is designed for:

- Water resources engineering students.
- Hydraulic and hydrology researchers.
- River modeling instructors.
- Field engineers who need a quick model setup preview.
- Early-stage professional users evaluating mobile hydraulic modeling workflows.

1.4 Main Capabilities

Capability	Description
Project Management	Create, open, review, and manage river modeling projects.
River Network Editing	Define river branches, nodes, junctions, and schematic river structure.
Cross-Section Input	Enter cross-section geometry, roughness values, and hydraulic properties.
Boundary Conditions	Define upstream and downstream conditions, including constant and time-series values.
Simulation Settings	Configure time step, duration, numerical scheme, and stability-related parameters.
Diagnostics	Check model readiness before running a simulation.
Results Review	View water level and discharge results through charts, tables, and exports.
Export	Export results and diagnostic summaries for further review.

2 WaRee11 v1.5 Workflow Overview

2.1 Summary-First Workflow

Version 1.5 introduces a summary-first workflow. Each major screen presents a concise overview first, then allows the user to open detailed screens only when needed.

Recommended Workflow

The recommended workflow is:

1. Open or create a project.
2. Review the Project Setup Progress card.
3. Define the river network.
4. Add cross-section geometry.
5. Configure boundary conditions.
6. Set simulation parameters.
7. Check model readiness.
8. Run the simulation.
9. Review and export results.

2.2 Five Main Application Areas

The v1.5 interface is organized around five main areas:

1. Project Preview / Project Launcher
2. River Network
3. Cross Sections
4. Simulation Settings
5. Results

2.3 Model Readiness Concept

Before running a simulation, WaRee11 checks whether required model components are complete. Readiness indicators help users identify missing or incomplete setup items.

Typical readiness checks include:

- Project metadata exists.
- River network contains at least one branch.
- Cross sections are assigned to the river network.
- Boundary conditions are defined.
- Simulation settings are valid.
- Stability checks are acceptable.

3 Project Preview and Project Launcher

3.1 Purpose of the Project Preview Screen

In WaRee11 v1.5, the Project Preview screen acts as the Project Launcher. It is not intended to be a full technical dashboard. Instead, it provides a clean entry point for opening recent projects, creating new projects, and checking the overall setup progress of the current project.

3.2 Main Components

The Project Preview screen should contain:

- Current Project card.
- Project Setup Progress card.
- Create New Project button.
- Recent Projects list.

3.3 Current Project Card

The Current Project card displays the active project name, location, last modified date, and a short status label.

Example information:

- Project name.
- River or basin name.
- Study area.
- Current setup status.
- Last opened date.

3.4 Project Setup Progress Card

The Project Setup Progress card summarizes how complete the project is. It should show:

- Large percentage value.
- Status label such as *Not Started*, *In Progress*, or *Ready*.
- Horizontal progress bar.
- Footer message such as *3 of 5 required items ready*.

3.5 Recent Projects

The Recent Projects list allows users to quickly reopen previous work. Each row may include:

- Project name.
- Study area.
- Last modified date.
- Setup status.

3.6 Opening Project Detail

Tapping a project row, the View Details button, or the progress card should open the Project Detail screen.

4 Project Detail Screen

4.1 Purpose

The Project Detail screen contains the complete technical and management information for a project. It separates detailed project information from the cleaner Project Preview screen.

4.2 Recommended Sections

The Project Detail screen may include:

- Project metadata.
- Model summary.
- Workflow entry points.
- Readiness and diagnostics summary.
- Project management actions.

4.3 Project Metadata

Project metadata may include:

Field	Description
Project Name	Name of the river modeling project.
River Name	Main river or channel being modeled.
Study Area	Geographic location or basin name.
Coordinate Reference	Optional GIS coordinate reference.
Created Date	Date when the project was created.
Modified Date	Date when the project was last edited.
Description	Short project description or modeling objective.

4.4 Model Summary

The model summary gives a technical overview of the current model setup:

- Number of river branches.
- Number of junctions.
- Number of cross sections.
- Boundary condition status.

- Simulation setting status.
- Latest simulation result status.

4.5 Project Management Actions

Possible actions include:

- Rename project.
- Duplicate project.
- Delete project.
- Export project summary.
- Open diagnostics report.

5 River Network Module

5.1 Purpose

The River Network module defines the structure of the river system. This includes river branches, nodes, junctions, and connections between upstream and downstream parts of the model.

5.2 River Network Concepts

A river network in WaRee11 may contain:

- **River Branch:** A connected river reach between upstream and downstream nodes.
- **Node:** A point representing an upstream end, downstream end, junction, or internal point.
- **Junction:** A connection point where multiple branches meet.
- **Chainage:** A distance coordinate along the river branch.

5.3 Creating a River Network

The typical steps are:

1. Open the River Network module.
2. Create a new branch.
3. Add upstream and downstream nodes.
4. Assign chainage or distance information.
5. Add junctions if required.
6. Save the network.

5.4 Network Summary

The v1.5 River Network screen should show a summary before details. Example summary metrics include:

- Number of branches.
- Number of nodes.
- Number of junctions.
- Network readiness status.

5.5 Common Network Problems

Problem	Suggested Action
No river branch defined	Add at least one branch before continuing.
Disconnected nodes	Check branch connections and junction links.
Missing chainage	Enter distance or chainage values for each required node.
Duplicate node names	Rename nodes to avoid confusion.
Invalid flow direction	Confirm upstream and downstream orientation.

6 Cross Section Module

6.1 Purpose

The Cross Section module defines the channel geometry used by the hydraulic solver. Cross-section data are essential for computing flow area, wetted perimeter, hydraulic radius, conveyance, water level, and discharge relationships.

6.2 Cross Section Data

A cross section may include:

- Cross-section name or ID.
- Chainage location.
- Station-elevation points.
- Manning roughness coefficient.
- Initial water level.
- Optional rating curve information.

6.3 Entering Cross Section Geometry

The user can enter cross-section points as station and elevation pairs.

Point	Station	Elevation
1	0.0	12.5
2	5.0	10.8
3	10.0	10.2
4	15.0	10.9
5	20.0	12.7

6.4 Manning Roughness

The Manning roughness coefficient controls resistance to flow. A higher Manning value represents greater resistance, such as vegetation, rough channel beds, or irregular banks.

The Manning equation is commonly written as:

$$Q = \frac{1}{n} AR^{2/3} S^{1/2} \quad (6.1)$$

where:

- Q = discharge
- n = Manning roughness coefficient
- A = flow area
- R = hydraulic radius
- S = energy slope or bed slope

6.5 Cross Section Readiness

A cross section is generally considered ready when:

- Geometry points are valid.
- At least two or more station-elevation points exist.
- Manning roughness is assigned.
- Chainage location is assigned.
- The section is connected to a river branch.

6.6 Rating Curve Preview

The rating curve preview shows the relationship between water level and discharge. It helps users check whether the cross-section geometry and roughness values produce reasonable hydraulic behavior.

7 Boundary Conditions

7.1 Purpose

Boundary conditions define how water enters and leaves the model domain. A one-dimensional river model normally requires upstream and downstream boundary conditions.

7.2 Boundary Condition Types

Type	Description
Upstream Discharge	Defines inflow discharge at the upstream boundary.
Upstream Water Level	Defines water level at the upstream boundary.
Downstream Water Level	Defines stage at the downstream boundary.
Downstream Discharge	Defines outflow discharge at the downstream boundary.
Time-Series Hydrograph	Defines time-varying discharge or water level.
Constant Value	Defines a fixed boundary value over the simulation period.

7.3 Creating a Boundary Condition

Typical steps:

1. Open the Boundary Conditions section from Simulation Settings or Project Detail.
2. Select upstream or downstream boundary.
3. Choose boundary condition type.
4. Enter constant value or time-series data.
5. Preview the hydrograph if time-series data are used.
6. Save the boundary condition.

7.4 Time-Series Boundary Conditions

A time-series boundary condition contains values that change over time.

Example:

Time	Discharge
0 min	50 m ³ /s

30 min	65 m ³ /s
60 min	90 m ³ /s
90 min	75 m ³ /s
120 min	55 m ³ /s

7.5 Boundary Condition Diagnostics

The application should check:

- Upstream boundary exists.
- Downstream boundary exists.
- Time-series data cover the simulation duration.
- Units are consistent.
- Values are physically reasonable.

8 Simulation Settings

8.1 Purpose

The Simulation Settings module defines how the hydraulic simulation will be executed.

8.2 Main Settings

Setting	Description
Time Step	Computational time interval.
Total Duration	Total simulation time.
Numerical Scheme	Solver method used by the model.
Courant Number	Stability-related value for explicit schemes.
Initial Water Level	Starting water level condition.
Output Interval	Time interval for saving results.

8.3 Numerical Scheme

WaRee11 may support several numerical schemes depending on the development version, such as:

- MacCormack scheme.
- Lax-Wendroff scheme.
- Finite-volume scheme.
- Rusanov flux.
- HLL flux.

8.4 Stability and CFL Condition

For explicit hydraulic solvers, the computational time step should satisfy a stability condition. A common form of the Courant-Friedrichs-Lewy condition is:

$$C = \frac{u\Delta t}{\Delta x} \quad (8.1)$$

where:

- C = Courant number
- u = flow velocity or wave speed

- Δt = time step
- Δx = spatial step

Important

If the time step is too large, the simulation may become unstable. Users should review the CFL warning before running the model.

8.5 Pre-Run Check

Before running a simulation, WaRee11 should check:

- River network readiness.
- Cross-section readiness.
- Boundary condition readiness.
- Simulation setting validity.
- Estimated stability condition.

9 Running a Simulation

9.1 Starting a Simulation

To run a simulation:

1. Open the Simulation Settings screen.
2. Confirm that all required items are ready.
3. Review diagnostics.
4. Tap Run Simulation.
5. Wait for the simulation process to complete.
6. Open the Results screen.

9.2 Simulation Status

During a simulation, the application may show:

- Running status.
- Current simulation time.
- Progress percentage.
- Warning messages.
- Completion message.

9.3 Failed Simulation

A simulation may fail because of:

- Missing boundary conditions.
- Invalid cross-section geometry.
- Unstable time step.
- Negative depth or unrealistic hydraulic state.
- Internal numerical error.

When a simulation fails, users should open the diagnostic report and correct the problem before running again.

10 Results Module

10.1 Purpose

The Results module displays computed hydraulic outputs such as water level, discharge, and diagnostic information.

10.2 Result Types

Possible result variables include:

- Water level.
- Discharge.
- Flow depth.
- Velocity.
- Diagnostic warnings.

10.3 Viewing Charts

Charts allow users to review changes over time or along the river. A typical result chart may show:

- Water level versus time.
- Discharge versus time.
- Water surface profile along the river.
- Comparison between scenarios.

10.4 Viewing Result Tables

The result table provides numerical values. Example table fields:

Time	Location	Water Level	Discharge
0 min	CS-01	10.20 m	50.0 m ³ /s
30 min	CS-01	10.45 m	65.0 m ³ /s
60 min	CS-01	10.80 m	90.0 m ³ /s

10.5 Scenario Comparison

If scenario management is enabled, users may compare results from multiple simulations. This is useful for checking the effect of:

- Different roughness values.
- Different inflow hydrographs.
- Different downstream water levels.
- Different time steps or numerical schemes.

11 Diagnostics and Model Readiness

11.1 Purpose

Diagnostics help users identify problems before and after simulation. This is especially important for students and early-stage model users.

11.2 Diagnostic Categories

Category	Example Checks
Project	Project name and metadata exist.
Network	Branches and nodes are connected.
Cross Sections	Geometry, roughness, and chainage are valid.
Boundary Conditions	Upstream and downstream conditions exist.
Simulation Settings	Time step, duration, and scheme are valid.
Stability	CFL estimate is acceptable.
Results	Simulation output exists and can be displayed.

11.3 Readiness Status

A readiness status may be shown as:

- **Not Started:** Required information has not been entered.
- **Incomplete:** Some information exists, but required items are missing.
- **Ready:** Required model setup is complete.
- **Warning:** The model can run, but the user should review potential problems.
- **Error:** The model cannot run until the issue is corrected.

11.4 Using Diagnostics for Learning

Diagnostics are not only error messages. They are also teaching tools. They help users understand why a hydraulic model requires complete geometry, boundary conditions, and stable numerical settings.

12 Exporting Data

12.1 Export Options

WaRee11 may support exports such as:

- Simulation result CSV.
- Diagnostic report CSV.
- Project summary report.
- Scenario comparison table.

12.2 Exporting Simulation Results

To export simulation results:

1. Open the Results screen.
2. Select the result variable.
3. Select time range or output interval if available.
4. Tap Export.
5. Choose the destination using the iPad share sheet.

12.3 Using Exported CSV Files

Exported CSV files can be opened in:

- Microsoft Excel.
- Numbers.
- Google Sheets.
- MATLAB.
- Python.
- R.
- GIS or external analysis software.

13 Example Workflow

13.1 Example: Simple One-Branch River Model

This example describes a basic model with one river branch, three cross sections, one upstream discharge boundary, and one downstream water level boundary.

13.2 Step 1: Create Project

Create a new project and enter:

- Project name: Simple River Example.
- River name: Example River.
- Study area: Training Basin.

13.3 Step 2: Define River Network

Create one river branch from upstream node U1 to downstream node D1.

13.4 Step 3: Add Cross Sections

Add three cross sections:

Cross Section	Chainage	Manning n
CS-01	0 m	0.030
CS-02	500 m	0.030
CS-03	1000 m	0.035

13.5 Step 4: Add Boundary Conditions

Use:

- Upstream discharge: 50 m³/s.
- Downstream water level: 10.0 m.

13.6 Step 5: Set Simulation Settings

Example settings:

Setting	Value
Time Step	10 seconds
Total Duration	2 hours
Scheme	Finite-volume or selected available scheme
Output Interval	5 minutes

13.7 Step 6: Run and Review

Run the simulation, then open the Results screen to review water level and discharge outputs.

14 Troubleshooting

14.1 Project Does Not Show as Ready

Check whether all required components are complete:

- River network exists.
- Cross sections are connected.
- Boundary conditions are defined.
- Simulation settings are valid.

14.2 Simulation Button is Disabled

The Run Simulation button may be disabled if the readiness check fails. Open the diagnostic summary to see missing items.

14.3 Results Are Empty

Possible causes:

- Simulation has not been run.
- Simulation failed.
- Output interval is not configured.
- Selected result variable has no data.

14.4 Unstable Simulation

If the simulation becomes unstable:

- Reduce the time step.
- Check cross-section geometry.
- Check boundary condition values.
- Review Manning roughness.
- Check for unrealistic initial water levels.

15 Glossary

Term	Definition
Boundary Condition	A required hydraulic condition at the edge of the model domain.
Chainage	Distance measured along a river branch.
Cross Section	Channel geometry at a specific river location.
Discharge	Flow rate, usually expressed as m^3/s .
Hydraulic Radius	Flow area divided by wetted perimeter.
Manning n	Roughness coefficient used in open-channel flow calculations.
River Branch	A connected reach of river between nodes.
Scenario	A saved model setup or simulation case used for comparison.
Time Step	Computational time interval used by the solver.
Water Level	Elevation of the water surface.

A Suggested Screenshot List

The final manual should include screenshots for the following screens:

1. WaRee11 launch screen.
2. Project Preview / Project Launcher.
3. Project Detail screen.
4. River Network summary.
5. River Network editor.
6. Cross Section summary.
7. Cross Section editor.
8. Rating curve preview.
9. Boundary Condition manager.
10. Simulation Settings screen.
11. Diagnostics report.
12. Results chart.
13. Results table.
14. Export screen or iPad share sheet.

B Suggested Future Manual Sections

Future versions of this manual may include:

- Full mathematical solver documentation.
- Finite-volume method explanation.
- Calibration workflow.
- Observed versus simulated comparison.
- GIS editing layer workflow.
- Classroom tutorial exercises.
- Field data collection workflow.

C Placeholder Figure Code

Use the following LaTeX structure when inserting real screenshots:

```
\begin{figure}[H]
  \centering
  \includegraphics[width=0.85\textwidth]{screenshots/project_launcher.png}
  \caption{Project Launcher screen in WaRee11 v1.5.}
  \label{fig:project-launcher}
\end{figure}
```


Closing Note

WaRee11 v1.5 is designed to make river hydraulic modeling more accessible on iPad. This manual should evolve together with the application, especially as the interface, solver, diagnostics, scenario management, calibration tools, and GIS editing layer become more mature.